[Final Year Project Proposal]

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Date (08 August 2025)

Project Details

Project Title	**Lumina: Deep Search Assistant Across All Extensions**		
Project Area of Specialization	Artificial Intelligence (AI), Machine Learning (ML), Natural Language Processing (NLP), Multimodal Information Retrieval, and Software Engineering.		
List Related Core Subjects	Data Structures and Algorithms (DSA) Database Systems (DB) Software Engineering (ISE) Human-Computer Interaction (HCI) Information Security (IS) Operating Systems (OS) Software Requirement Engineering (SRE) Software Construction & Development (SCD) Software Design & Architecture (SDA) Web Engineering(WE)		
List Related Elective Subjects	Fundamental of Management (FOM) Artificial Intelligence (AI) Machine Learning (ML)		
Project Start Date	2025-08-02	**Project End Date**	2026-07-02

Project Summary (less than 2500 characters)	The project pertains to development of **Lumina: Deep Search Assistant Across All Extensions** that enables users to search, retrieve, and understand knowledge hidden inside their personal files across multiple formats — text, images, audio, video, and structured data. Unlike traditional systems such as Google or Windows Search, **Lumina** not only locates files but also analyzes their contents, summarizes them, and presents context-aware insights. The system integrates advanced AI models for each modality: MiniLM for text embeddings, CLIP for images and video frames, Whisper for speech-to-text transcription, and FAISS for efficient vector search. It supports multimodal queries, allowing users to search with text, voice, or images, and employs hybrid retrieval — combining keyword-based precision with semantic understanding. Users interact through a chatbot-style interface that can generate summaries, explanations, and knowledge timelines with timestamps (e.g., identifying the exact moment a topic is explained in a lecture recording). To improve accuracy in real-world usage, the project includes fine-tuning of CLIP and Whisper models. CLIP will be adapted for specialized domains such as lecture slides or medical images, while Whisper will be fine-tuned for Urdu-English mixed speech and local accents, significantly improving transcription accuracy. Lightweight LoRA fine-tuning methods make this feasible on standard hardware. The proposed system architecture includes a desktop-based interface (React/Electron) and a Python backend (FastAPI/Flask), ensuring cross-platform usability and complete privacy since all processing remains local with no cloud dependency. Core features include multimodal search, multilingual support, knowledge graph construction to link related files, and task-based automation such as summarization, fact extraction, and report generation. This project's area of specialization lies in **Artificial Intelligence, Machine Learning, Natural Language Processing, Information Retrieval, and Software Engineering**. Its practical impact extends to students, researchers, doctors, lawyers, and organizations who manage large volumes of knowledge but lack intelligent tools to organize and extract insights. In short, the project delivers an innovative, privacy-preserving AI system that makes local knowledge search smarter, faster, and more meaningful.
Project Objectives (less than 2500 characters)	The main objective of this project is to develop **Lumina: Deep Search Assistant Across All Extensions** that enables intelligent, privacy-preserving search and understanding across a user's personal files, including text, images, audio, video, and structured data. Unlike traditional search systems, **Lumina** aims to go beyond keyword matching and provide semantic, context-aware, and summarized results that make knowledge easier to find and understand. Specific objectives include: 1. **Multimodal Search & Retrieval** o Enable search across multiple file types (documents, images, audio, video, spreadsheets). o Support queries in text, image, or audio format for flexible interaction.

	2. **Semantic & Hybrid Search** ○ Implement semantic search to retrieve conceptually related results (e.g., "fever" → "infection, virus"). ○ Combine semantic search with keyword-based retrieval for both precision and context. 3. **Summarization & Question Answering** ○ Provide summaries of lecture notes, meeting recordings, or reports. ○ Support chatbot-style Q&A where users can ask natural language questions about their files. 4. **Knowledge Timeline & Contextual Navigation** ○ Display results from lectures or meetings with timestamp-based navigation. ○ Help users locate the exact point where a concept was explained. 5. **Model Fine-Tuning for Local Needs** ○ Fine-tune Whisper for Urdu-English mixed speech, local accents, and technical jargon. ○ Fine-tune CLIP for specialized images like lecture slides or X-rays. ○ Use lightweight LoRA methods to ensure training can be done on modest hardware. 6. **Task-Based Automation** ○ Automate tasks such as extracting structured facts (e.g., patient details, key topics) or generating reports. 7. **Multilingual & Cross-Lingual Support** ○ Allow users to query and retrieve knowledge in multiple languages (e.g., English, Urdu). 8. **Privacy & Local Deployment** ○ Ensure all processing is done on the user's device with no dependency on cloud services. ○ Provide a cross-platform desktop interface (React/Electron) with a Python backend (FastAPI/Flask). Through these objectives, the project aims to deliver a system that is **accurate, private, user-friendly, and practically useful** for students, researchers, doctors, lawyers, and organizations handling large volumes of diverse information.
Project Implementation Method (less than 2500 characters)	The implementation of **Lumina: Deep Search Assistant Across All Extensions** will be carried out in a step-by-step, modular approach to ensure smooth development, testing, and integration of all features. **Step 1: Core Text Search** • Begin with text-based search using pre-trained MiniLM embeddings for document representation. • Store embeddings in FAISS for fast similarity search. • Develop a simple query interface to test basic text retrieval. **Step 2: Image and Video Integration**

- Integrate CLIP to enable text-to-image and image-to-text search.
- For video, extract keyframes and process them with CLIP, while also extracting the audio track for Whisper-based transcription.
- Store embeddings in FAISS for cross-modal retrieval.

Step 3: Audio Processing

- Use Whisper Small for speech-to-text transcription with timestamps.
- Enable search across transcribed lecture or meeting audio files.
- Integrate CLAP for non-speech audio if required.

Step 4: Summarization and Q&A

- Connect the retrieval pipeline with a local small LLM (Mistral 7B or LLaMA 3).
- Implement summarization of documents, lecture notes, or meeting transcripts.
- Add a chatbot-style Q&A interface where users can ask questions in natural language.

Step 5: Knowledge Timeline and Visualization

- Develop timeline-based navigation for lectures and meetings, showing when a topic was explained.
- Implement knowledge graph linking related files (e.g., slides, notes, and lecture audio).

Step 6: Model Fine-Tuning

- Fine-tune Whisper on Urdu-English mixed speech and local accents using LoRA.
- Fine-tune CLIP with domain-specific datasets (e.g., lecture slides + notes, X-rays + reports).
- Evaluate performance improvements in transcription accuracy and cross-modal retrieval.

Step 7: Multilingual and Hybrid Search

- Add multilingual support using cross-lingual embeddings.
- Implement hybrid search that combines semantic and keyword retrieval for improved accuracy.

Step 8: Frontend and User Interface

- Build a desktop interface using React/Electron with search options for text, image, audio, and video queries.
- Provide chatbot-style interactions, mixed query inputs, and timeline visualization.
- Ensure a user-friendly, cross-platform design.

Step 9: Privacy and Deployment

- Ensure all components (embeddings, models, databases) run locally on the user's device with no cloud dependency.
- Package the system for cross-platform desktop deployment.

By following this phased implementation method, **Lumina** will evolve from a simple text-based search tool into a full-featured multimodal knowledge assistant capable of intelligent retrieval, summarization, and task automation while ensuring complete privacy.

Benefits of the Project (less than 2500 characters)	The development of **Lumina: Deep Search Assistant Across All Extensions** offers significant academic, professional, and practical benefits. 1. **Enhanced Productivity** ○ Saves time by allowing users to quickly locate information across documents, images, audio, and video. ○ Provides instant summaries and answers, reducing manual effort in reviewing large content. 2. **Cross-Modal Intelligence** ○ Enables users to search in one format and retrieve results in another (e.g., find the slide that matches a spoken explanation). ○ Improves knowledge accessibility for students, researchers, and professionals. 3. **Privacy and Security** ○ Runs completely offline, ensuring user data remains secure and private. ○ Eliminates dependency on third-party cloud services. 4. **Educational Benefits** ○ Students can revisit lectures using transcripts, summaries, and topic-based timelines. ○ Helps in exam preparation, note-taking, and quick concept revision. 5. **Research and Industry Use** ○ Researchers can organize large datasets (papers, lecture recordings, experiment notes). ○ Professionals can manage meetings, presentations, and training sessions effectively. 6. **Multilingual Support** ○ Provides accessibility in English, Urdu, and mixed text/speech, making it highly usable in Pakistan and other multilingual environments. 7. **Scalability and Extendibility** ○ Can be extended to support enterprise document management, medical reports, legal archives, and organizational knowledge bases. ○ Serves as a foundation for advanced applications such as intelligent tutoring systems and private enterprise assistants. 8. **Skill Development for Students (Team Benefit)** ○ Builds expertise in modern AI techniques like embeddings, multimodal search, speech recognition, and fine-tuning. ○ Strengthens software engineering skills through full-stack development, optimization, and privacy-preserving design. **Overall Benefit** This project empowers users with a **personal, intelligent, and secure knowledge assistant** that enhances learning, research, and productivity while addressing privacy concerns often overlooked in commercial solutions.

Technical Details of Final Deliverable (less than 2500 characters)	The final deliverable of **Lumina: Deep Search Assistant Across All Extensions** will be a **cross-platform desktop application** designed with a modular architecture to ensure performance, scalability, and privacy. **Core Components** 1. **Data Storage & Indexing** ○ All files (text, images, audio, video) will be embedded using specialized models. ○ Vector storage and fast similarity search will be implemented using **FAISS**. ○ Metadata will be managed via a lightweight database (SQLite/PostgreSQL). 2. **Models & Processing** ○ **Text Search** → MiniLM or Sentence Transformers for semantic embeddings. ○ **Image/Video Search** → CLIP for cross-modal embedding of images and keyframes. ○ **Audio Processing** → Whisper for transcription (speech-to-text). ○ **Summarization & Q&A** → Local LLM (Mistral 7B / LLaMA 3) for context-based answers. ○ **Fine-tuning** → LoRA-based tuning for Urdu-English mixed speech and domain-specific datasets. 3. **Functional Modules** ○ **Search Engine** → Hybrid semantic + keyword retrieval across multiple file formats. ○ **Multimodal Query Support** → Users can search by text, image, audio, or video. ○ **Knowledge Timeline** → Timeline navigation for lectures/meetings with synchronized transcripts. ○ **Knowledge Graph** → Visualization of related topics and files. ○ **Summarization & Notes** → On-demand summaries of large files, lectures, or videos. 4. **Frontend (User Interface)** ○ Built using **React + Electron** for a cross-platform desktop app (Windows/Linux/macOS). ○ Clean UI with separate buttons for **Text, Image, Audio, and Video search modes**. ○ Chatbot-style interaction for natural Q&A. ○ Visualization features (timeline, knowledge graph) for better navigation. 5. **Backend** ○ Implemented in **Python**, handling embeddings, search, and LLM responses. ○ API endpoints exposed via **FastAPI** to connect the backend with the desktop frontend. 6. **Privacy & Deployment** ○ All models and databases run **locally on the user's device**, ensuring no cloud dependency.

	○ Final deliverable packaged as a **standalone installer** for different operating systems. **Expected Output** • A working **desktop knowledge assistant** capable of intelligent multimodal search, summarization, and Q&A. • Source code, user documentation, and performance evaluation reports.
Final Deliverable of the Project	**Final Deliverable of the Project** The final deliverable of **Lumina: A Private Multimodal Knowledge Assistant** will be a fully functional **cross-platform desktop application** that provides: 1. **Multimodal Search** ○ Intelligent retrieval across text (PDF, DOCX, TXT), images, audio, and video. ○ Hybrid search combining semantic similarity and keyword matching. 2. **Multimodal Query Options** ○ Dedicated search modes for text, image, audio, and video queries. ○ Cross-modal retrieval (e.g., search by image → get relevant text/audio/video). 3. **Transcription & Summarization** ○ Automatic speech-to-text transcription for lectures/meetings. ○ Summaries of documents, videos, and audio recordings. 4. **Knowledge Navigation Tools** ○ Timeline-based playback for lectures/meetings. ○ Knowledge graph linking related content across different file types. 5. **Interactive Assistant** ○ Chatbot-style interface for Q&A based on user data. ○ Multilingual support (English, Urdu, and mixed text/speech). 6. **Privacy & Security** ○ Runs offline, ensuring no data leaves the user's device. ○ Packaged as a standalone installer for **Windows, Linux, and macOS**. **Deliverable Package Includes** • Cross-platform desktop application. • Source code with modular architecture. • Technical documentation and deployment guide. • User manual with installation and usage instructions. • Evaluation report (performance, accuracy, scalability).
Type of Industry	Information Technology (AI-powered Knowledge Management and Educational Technology)

Technologies	**Technologies** • **Programming Languages**: Python (backend, ML integration), JavaScript (React + Electron for frontend). • **Frameworks & Libraries**: ○ React, Electron → cross-platform desktop application. ○ FastAPI → backend API services. ○ FAISS → vector search and similarity retrieval. • **Machine Learning / AI Models**: ○ Sentence Transformers / MiniLM → text embeddings. ○ CLIP → cross-modal (image–text) embeddings. ○ Whisper → speech-to-text transcription. ○ Local LLMs (Mistral 7B, LLaMA 3) → summarization & Q&A. • **Fine-Tuning Approaches**: ○ **LoRA (Low-Rank Adaptation)** for efficient fine-tuning. ○ Whisper fine-tuned on Urdu-English mixed speech. ○ CLIP fine-tuned on domain-specific datasets (e.g., lecture slides + notes). • **Databases**: SQLite / PostgreSQL (for metadata & indexing). • **Deployment & Packaging**: Electron Builder (cross-platform desktop installer). • **Visualization Tools**: D3.js / React libraries (timeline & knowledge graph visualization).
Sustainable Development Goals	**Sustainable Development Goals (SDGs) Addressed by the Project** 1. **SDG 4 – Quality Education** ○ Supports students and educators by providing intelligent search, lecture transcription, summaries, and multilingual support. ○ Enhances access to knowledge, especially for learners with limited resources. 2. **SDG 9 – Industry, Innovation, and Infrastructure** ○ Promotes innovation in AI-powered knowledge management. ○ Provides scalable, privacy-preserving infrastructure for managing educational and professional content. 3. **SDG 16 – Peace, Justice, and Strong Institutions** *(Maybe in future goals)* ○ Ensures privacy and security by keeping all data local, empowering individuals and institutions to manage their knowledge without surveillance or cloud dependency.

Project Key Milestones

Elapsed time in (days or weeks or month or quarter) since start of the project	Milestone	Deliverable
Month 1–2	**Requirement Analysis**	Finalized project requirements, dataset collection, and architecture design.
Month 3–4	**Prototype Development**	Basic multimodal search (text & document search) working prototype.
Month 5–6	**Model Integration**	Integration of Whisper (speech-to-text), CLIP (image-text search), and LLM for Q&A.
Month 7–8	**Fine-Tuning & Testing**	Fine-tuned models for Urdu-English and domain-specific datasets; functional testing.
Month 9–10	**UI/UX Implementation**	Desktop app (React + Electron) with interactive dashboard and chatbot UI.
Month 11	**System Integration**	Fully integrated multimodal assistant with search, summarization, and timeline.
Month 12	**Final Testing & Report**	Final deliverable, documentation, evaluation report, and deployment package.

Project Equipment Details

Item Name	Type	No. of Units	Per Unit Cost (in Rs)	Total (in Rs)
Development Laptop (for model training)	Equipment	1	160,000	160,000
External Storage / SSD (datasets backup)	Equipment	1	10,000	10,000
GPU Cloud Credits (optional fine-tuning)	Software/Cloud	-	30,000	30,000
Miscellaneous (electricity, internet)	Miscellaneous	1	10,000	10,000
			Total in (Rs)	**240,000**